

NC lab individual study

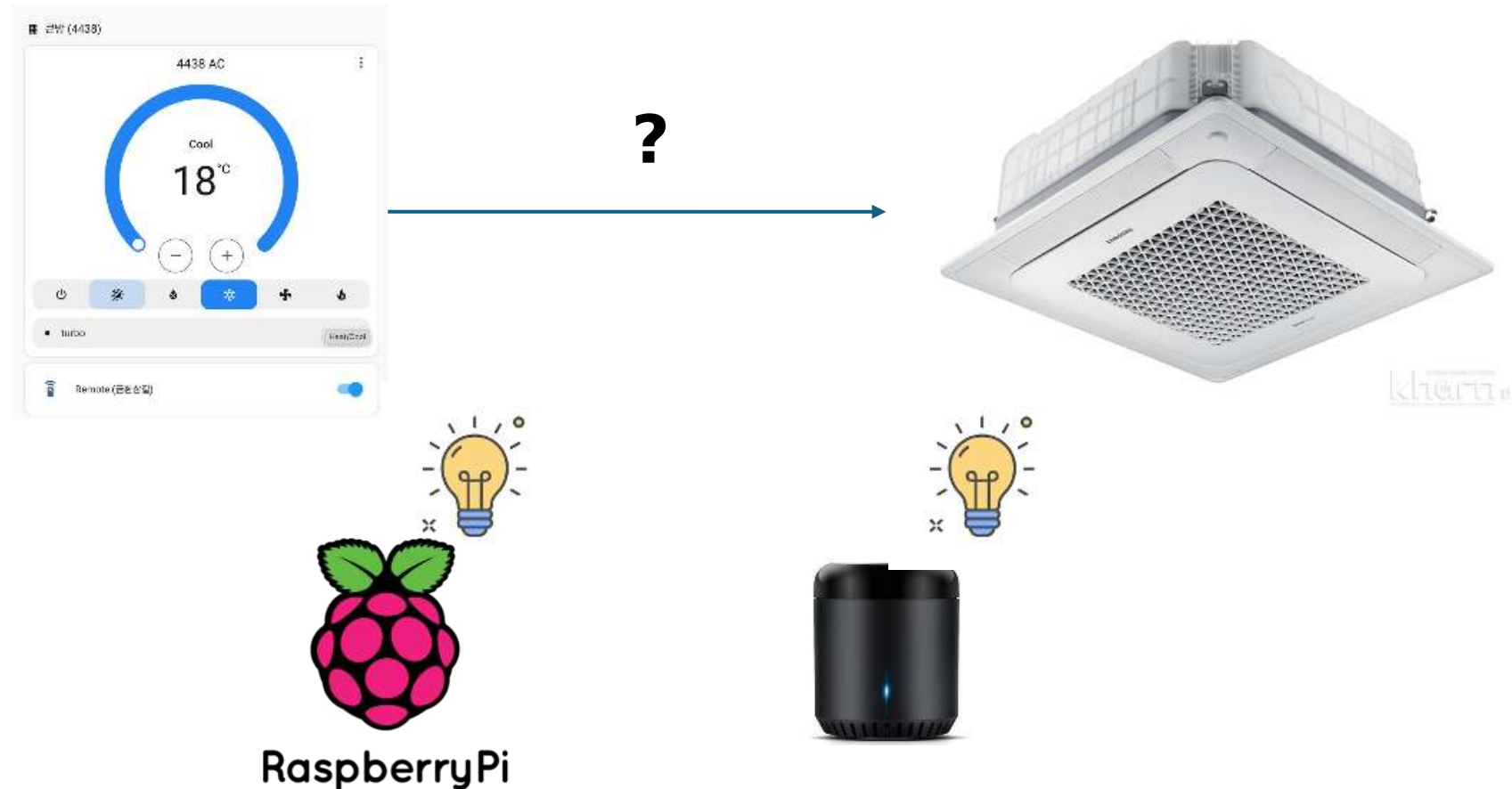
20200690 Euseok Han

Content

1. Overall experience in individual study
 1. Enjoyment of understanding the system
 2. What I did: AudioSep, Hivemind
2. My research – Hivemind Auth Server
3. What I learned & Future works

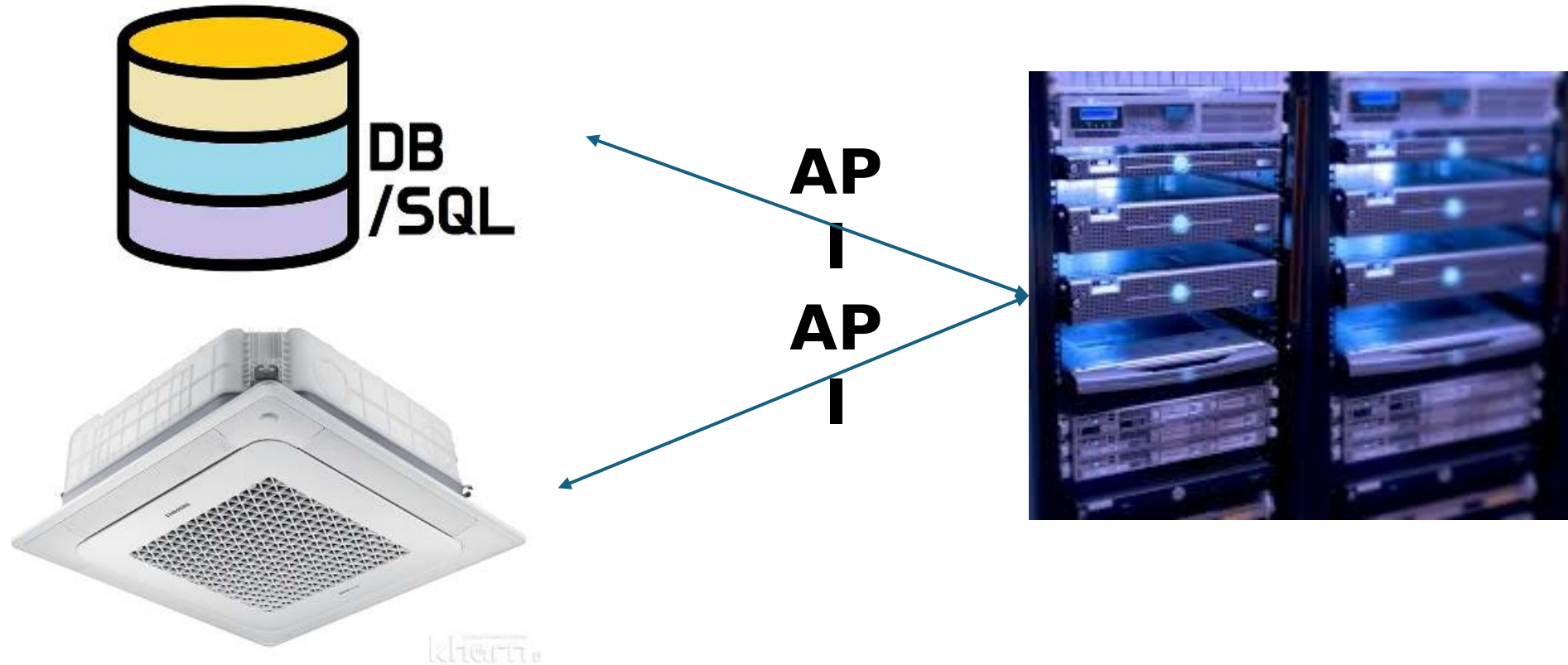
1-1 Enjoyment of understanding the system

- Step by step learning



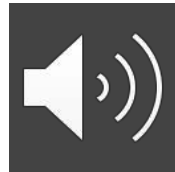
1-1 Enjoyment of understanding the system

- Interesting

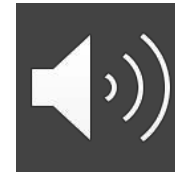


1-2 What I did - AudioSep

- Separating audio to detect event in the space
- Analyzed result with matlab



Coffee grinder with
noise



Separated coffee grinder
sound

1-2 What I did - Hivemind

Hivemind

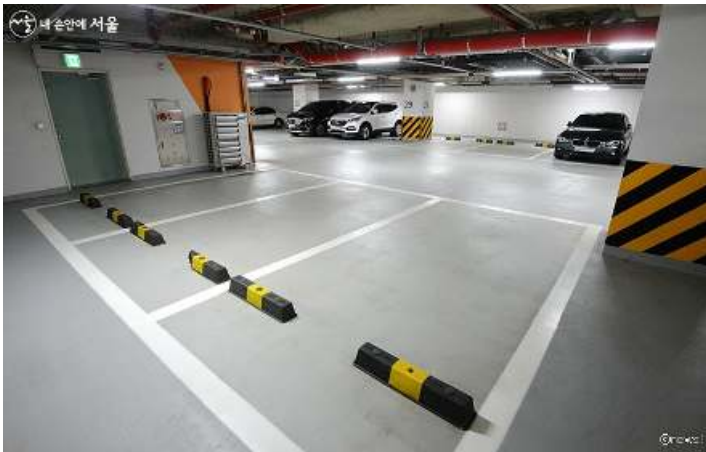
- Resolve conflict...

What is future for Hivemind?

- Raspberry pi in every room is too costly
- How to enhance participation while lowering cost

2. Hivemind authenticator – motivation (Alternative **deployment** strategy)

No matter the space, an admin always exists.



Resolve conflict(Calibration) and fade out

- Temporary deployment → Active participation is more important!

Two requirements of such system

Anyone currently using the space should be able to access the system **easily**



(1) Any users in the current space



Hotels.com™



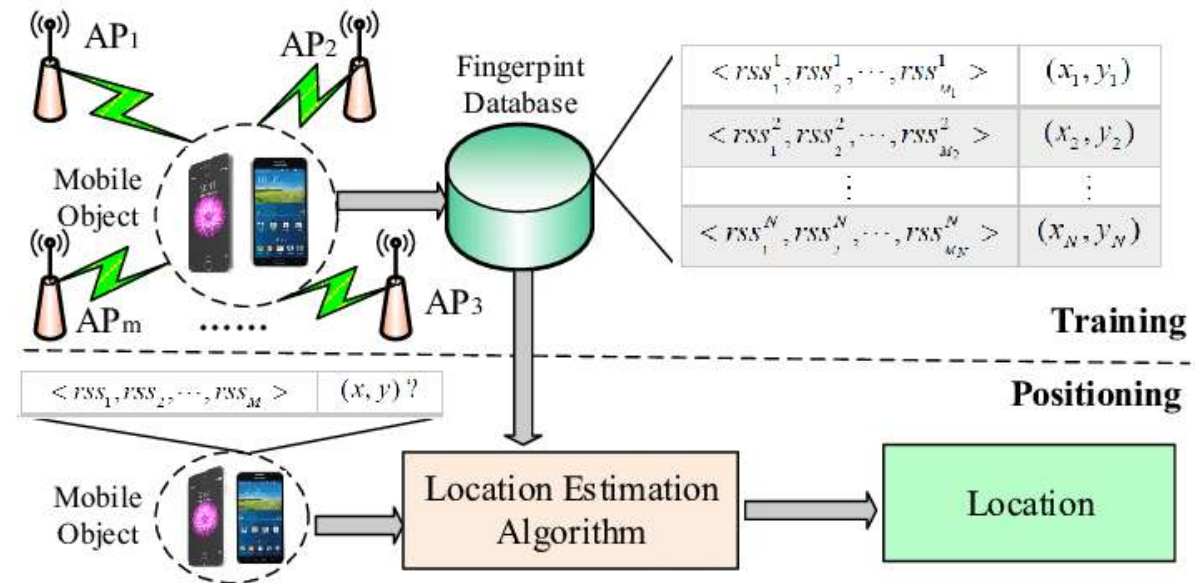
(2) Easy access

Approach: QR code + Geolocation API



- URL + QR code
- Did he/she enter?

Actually arrive at the place?

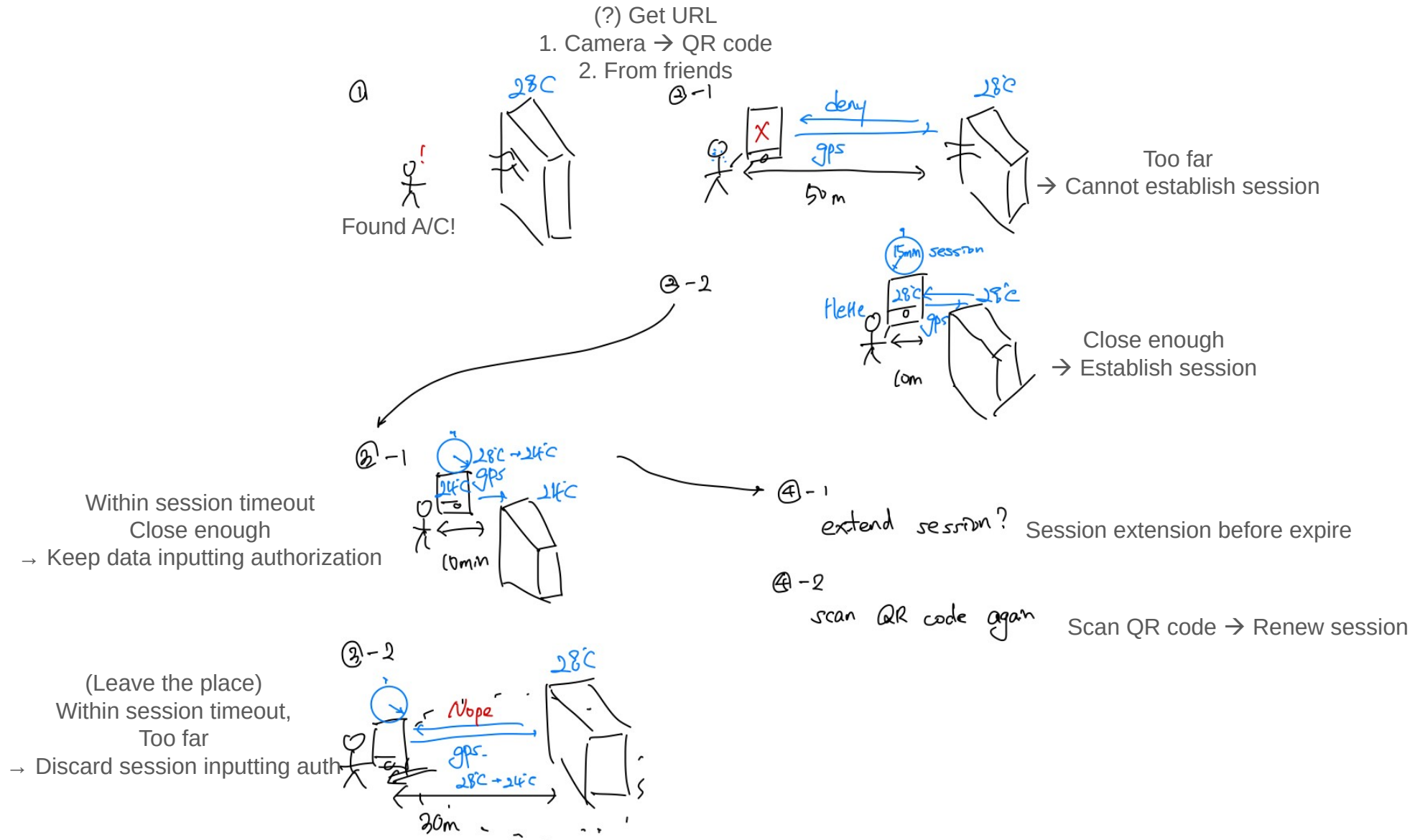


Geolocation API
(supported by major browsers)
(GPS + cell tower + wifi fingerprint
+ ...)

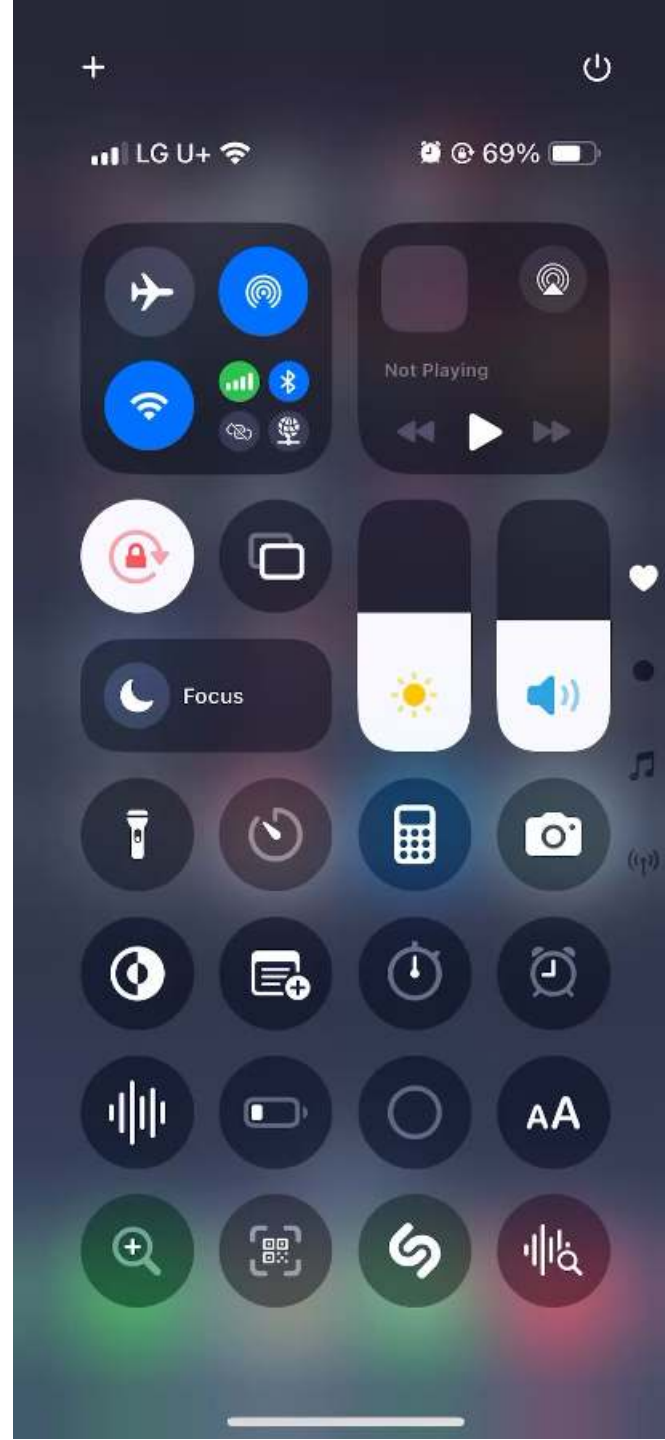
- Does he/she still here?

Still at the place?

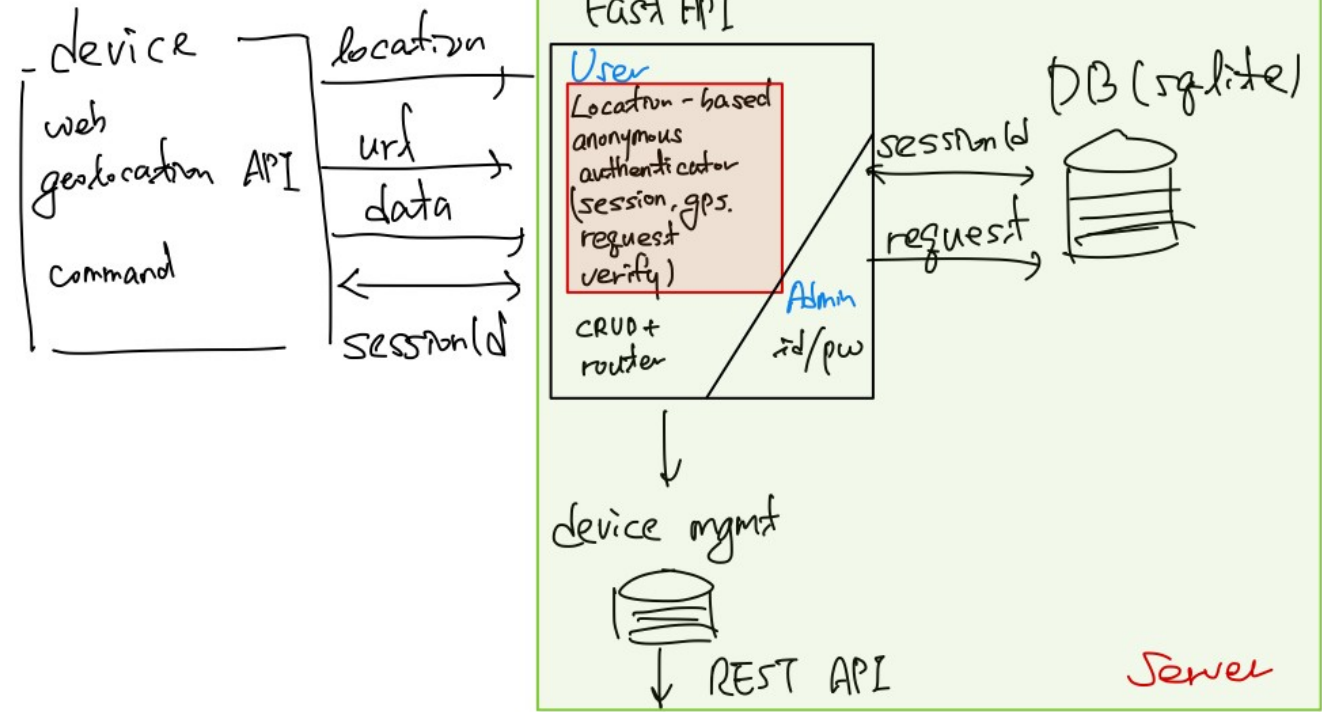
System Walkthrough



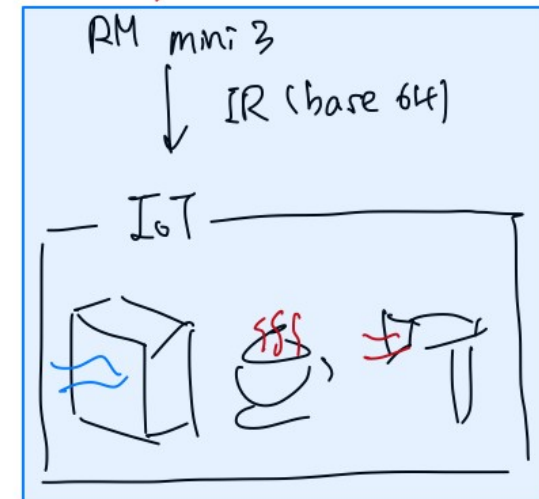
Demo



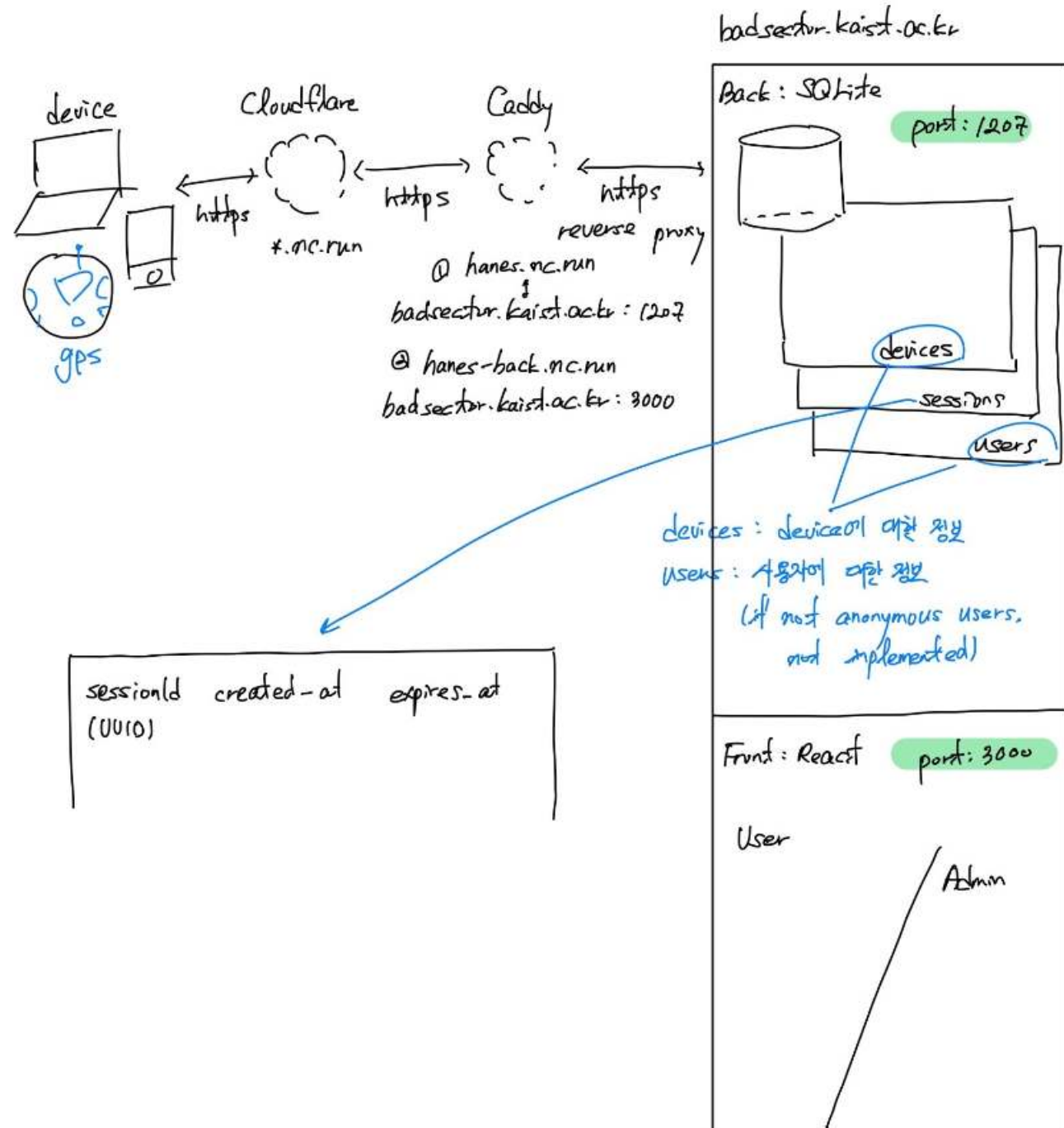
System Architecture



Not implemented...



Implementation



Operation flow

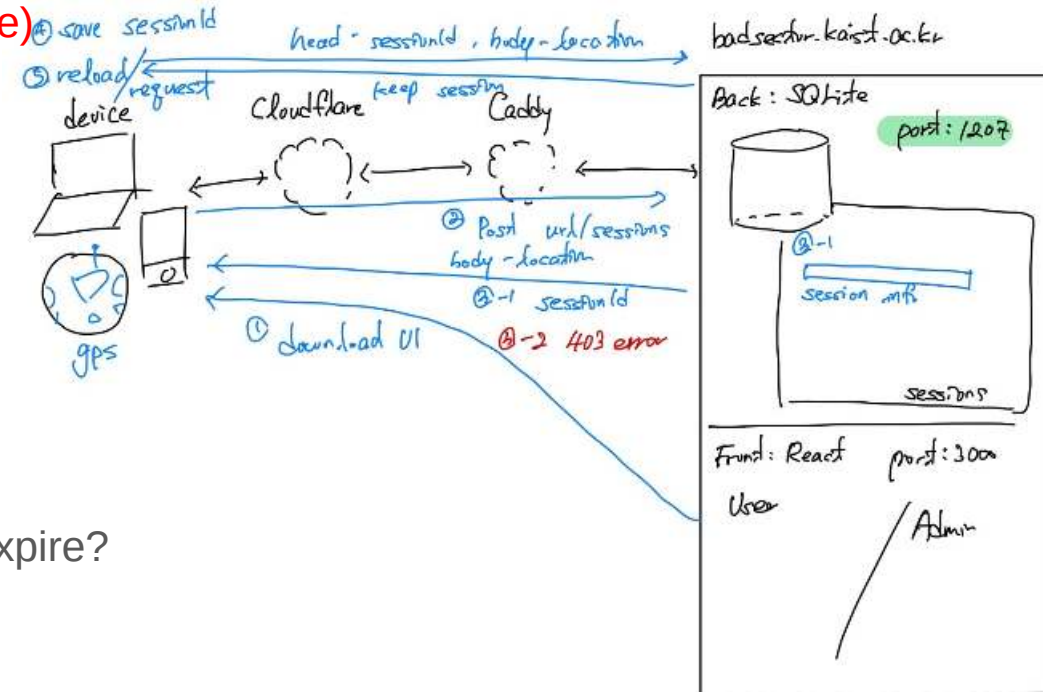
1. user scan QR code(<https://hanes.nc.run>)
2. user download UI from <https://badsector.kaist.ac.kr:3000> (<https://hanes.nc.run/> - user page, <https://hanes.nc.run/admin> - admin)
3. UI automatically tries to get sessionId. But not exist initially → user/[user.js](#) calls `getOrCreateSession()`(defined in user/[sessionService.js](#))
4. `getOrCreateSession()` post <https://hanes-back.nc.run/sessions/> with location information in the body
 1. if the location is close enough, save session info in the DB
Return sessionId(**device only keeps sessionID in localStorage**)
 2. if the location is not close enough,
fastAPI raise HTTPException(403 error)
5. When the device reload the webpage or request, it sends its sessionId in the head,
location in the body
6. fastAPI validates sessionId / handles request

What is not implemented:

How to handle when the session expires?

What if the device escape from the boundary while session doesn't expire?

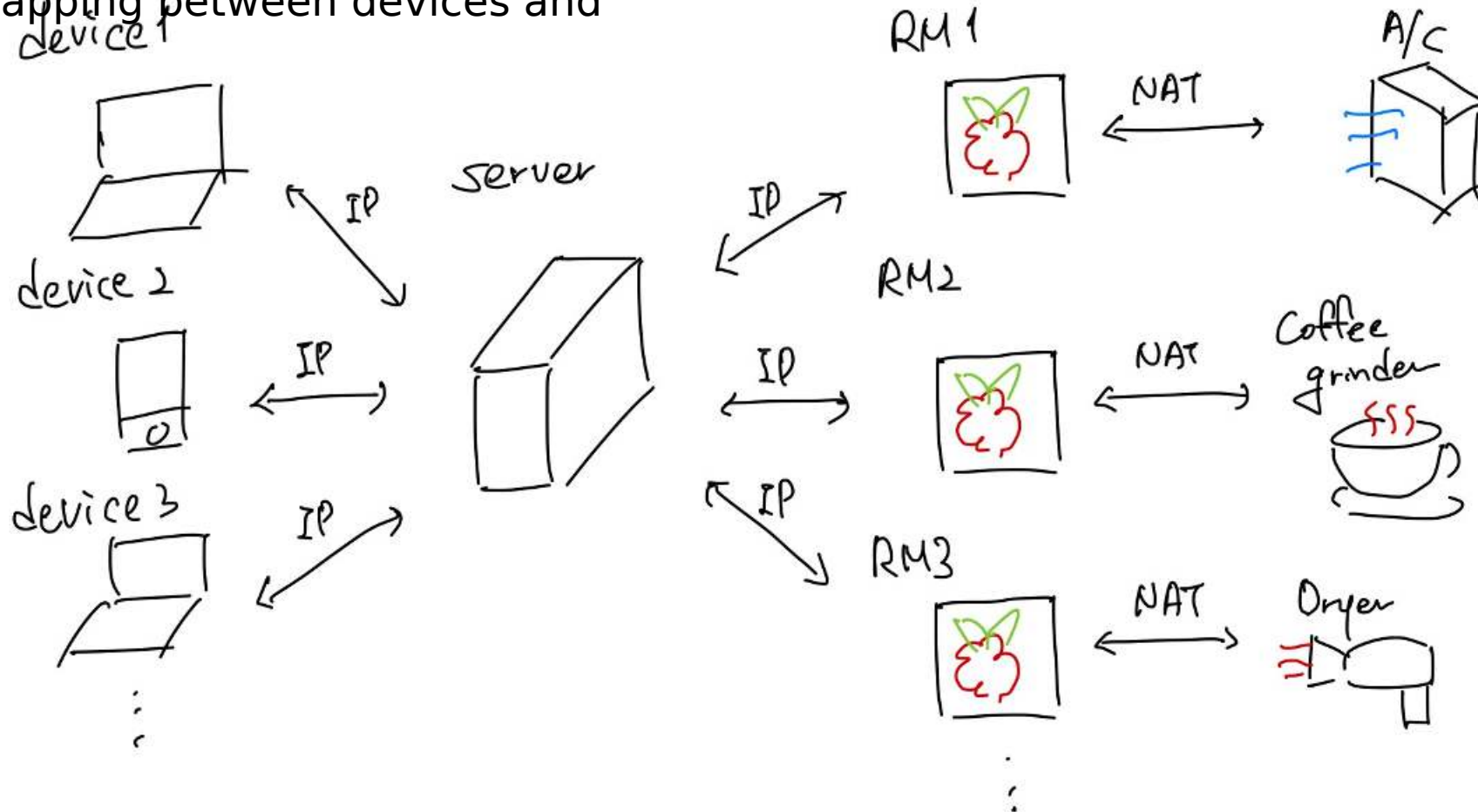
Rules for session time and managing session data



Platform architecture (example)

Can maintain multiple rooms with a server!

N-to-M mapping between devices and rooms



Possible extension



Where to build parking lot?

- Plate as id number
(not anonymous)



Any welfare facility?



Office temperature?

- Between coworkers
(acquaintances)

3. What I learned

Socially-appropriate Use of Noise-making Everyday Appliances in High-density Urban Housing

Anonymous Author(s)

Abstract

In high-density urban housing of modern cities, such as dormitories and shared flats, noises from everyday appliances often cause severe stress and even conflict within a community. This paper presents Social Appropriateness Guardian, a novel mobile-IoT system that facilitates appliance users' awareness of potential noise annoyance induced by an appliance. It employs an *in-the-wild* online sampling and prediction approach; it captures events of appliance use, which naturally and repeatedly occur on a daily basis, and samples its noise propagation at the moment of appliance use, leveraging nearby mobile and IoT devices. Furthermore, it presents *annoyance moderation services* as a new type of service built upon the SAG system. We deployed the SAG system along with annoyance moderation services in real-world high-density housing for two weeks and evaluated their usefulness and user acceptance.

1 Introduction

Home. It is a comforting place: calm, quiet, and cozy. Home is your own tranquil space where you detach from the outside world and relax after a long tiring day. But is it so? Houses in cities, they are quite different from this image of home. School dormitories, that we live in or have lived in, are surprisingly bustling and even wild. Noise from everywhere haunts us: washing machines, coffee grinders, parties, people stomping around, and vacuum cleaners. This noise frustrates

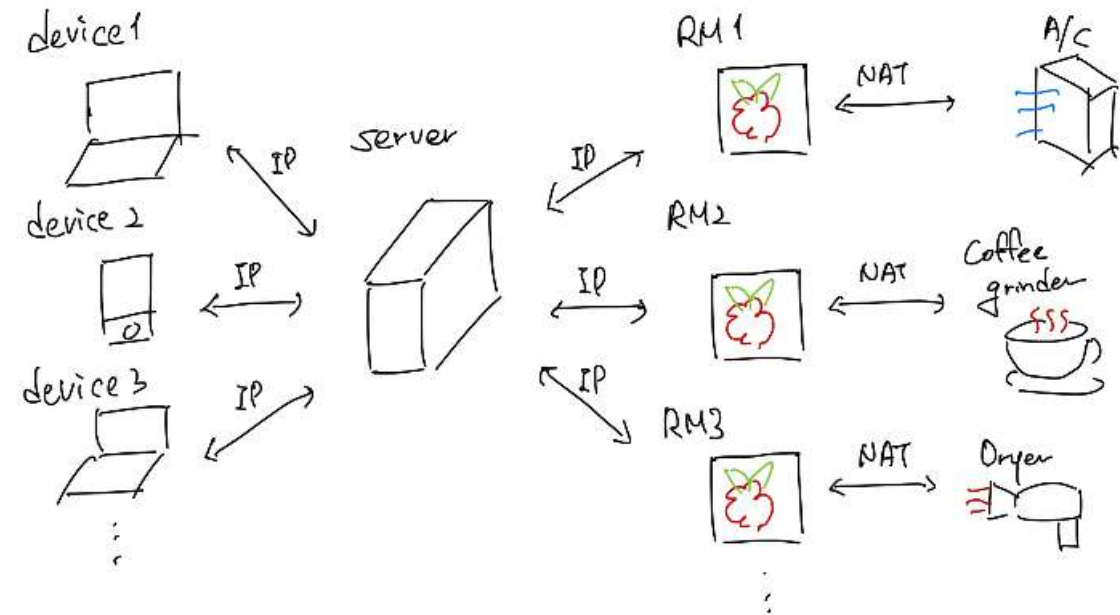


Figure 1: Socially-appropriate use of appliances

of nowhere, any time. Knocking on the wall of the next room or going upstairs is a temporary solution. In high-density housing, relaxation cannot be achieved alone.

We point out that a key problem is the lack of awareness of appliance noise propagation in buildings, leading to unwitting noise nuisance. An appliance user does not know how much the noise is disturbing neighbors. Similarly, it is not easy for a resident who suddenly hears a noise to identify the source and location of the noise. This adds ambiguity to the rules and regulations, making them difficult to comply with. For example, we all know that it is desirable to be quiet at home, but we do not know whether we should not run the vacuum cleaner at all, or whether we can run it after closing the door. As a result, even the most basic rule of communal living, "be quiet at night", is difficult to practice.

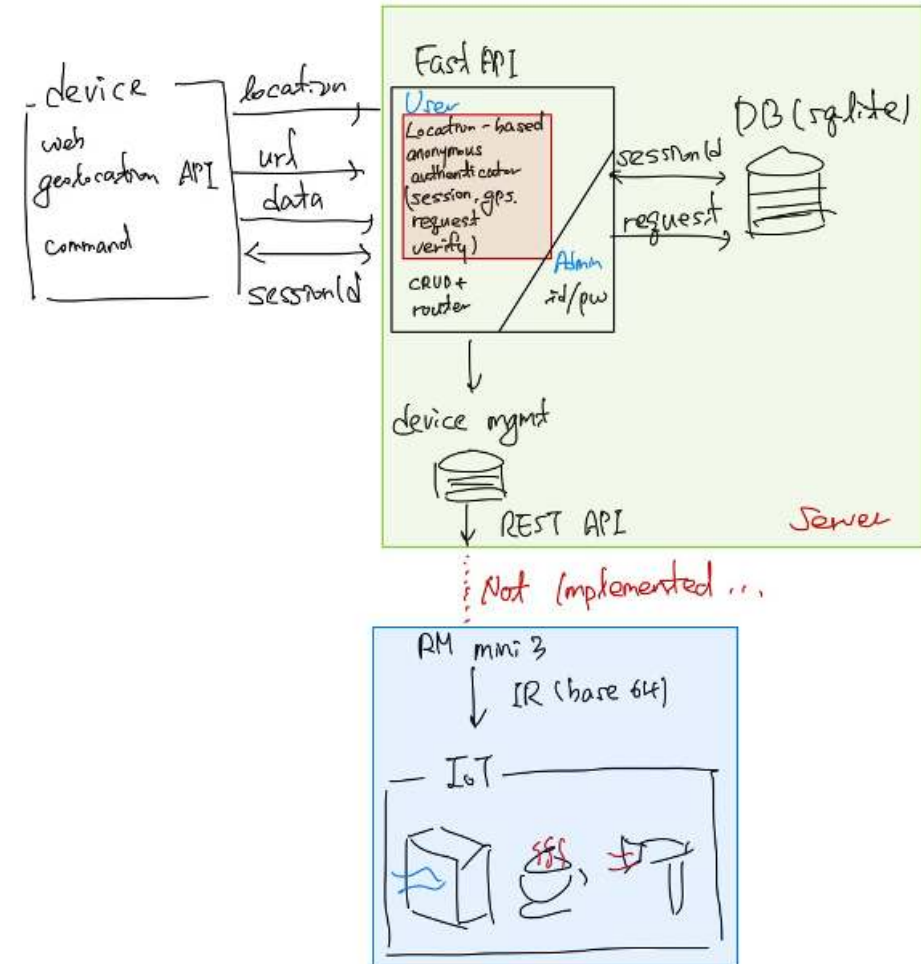
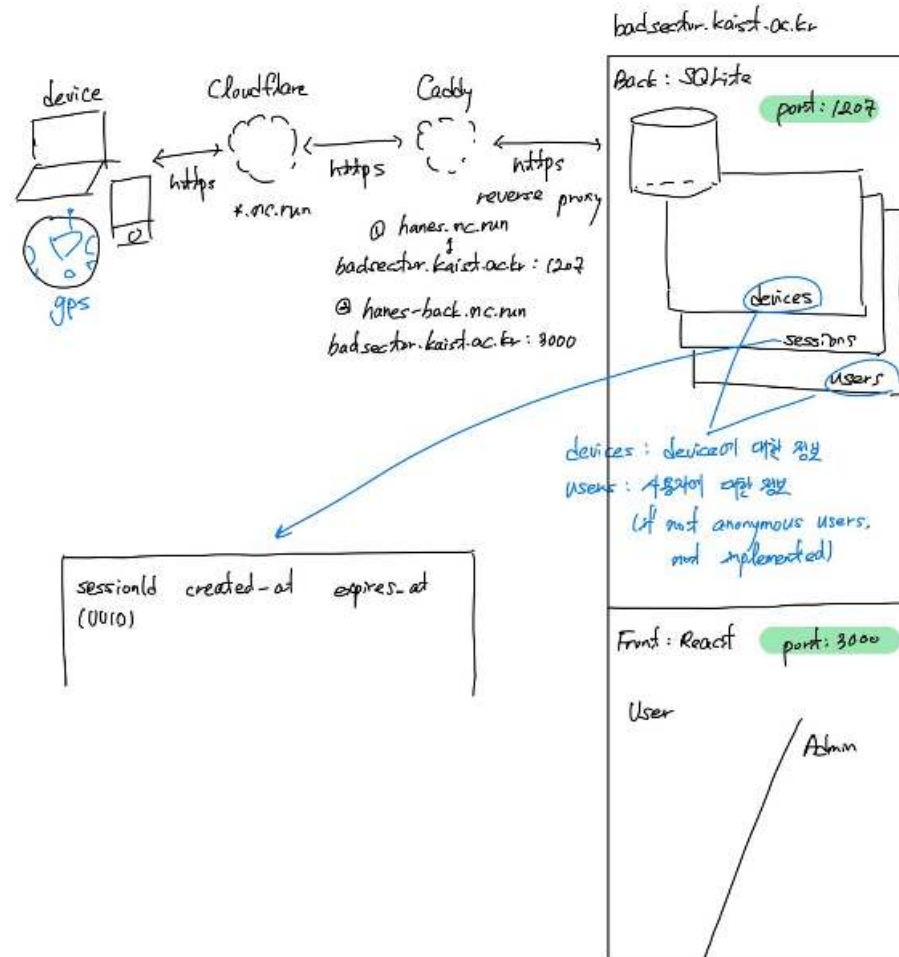
We present Social Appropriateness Guardian, a novel system that provides the awareness of noise annoyance to neigh-



How research
conducted

Fundamentals of Computer
Science
– Servers and Devices

3. What I learned



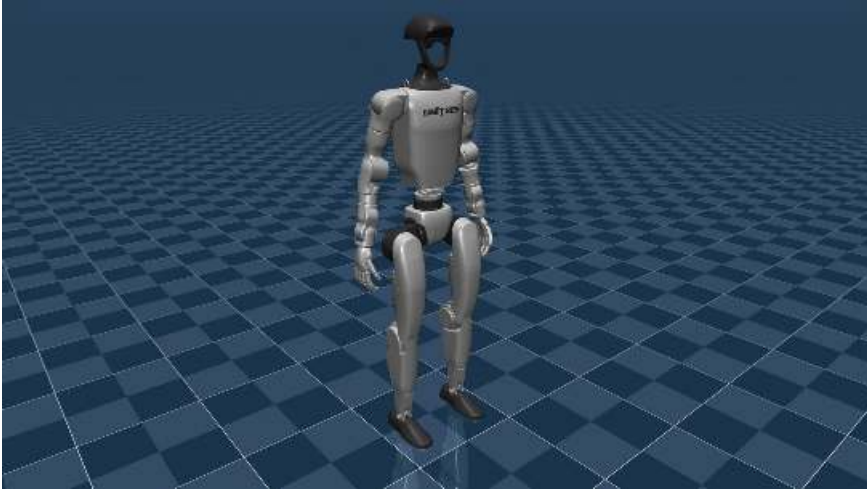
3. What I learned



[JavaScript (Run browser)] <-> [React (FrontEnd UI Library)]
<-> [FastAPI (BackEnd API Server)] <-> [SQLite (DB)]

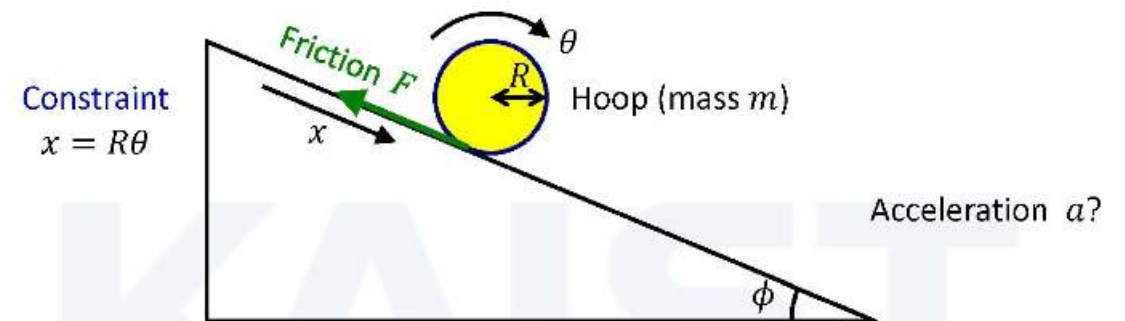


3. Future works



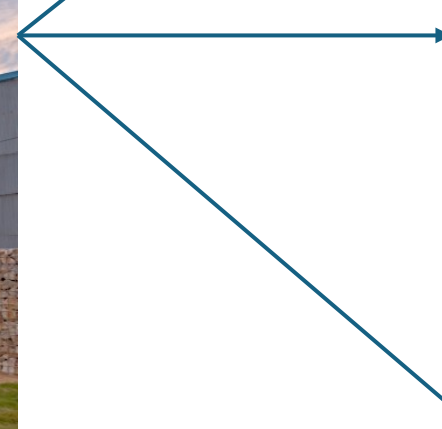
▼ safety-starter-agents	
> data	
▼ safe_rl	
> __pycache__	
▼ pg	
> __pycache__	
🔗 agents.py	
🔗 algos.py	
🔗 buffer.py	
🔗 network.py	M
🔗 run_agent.py	1, M
🔗 trust_region.py	M
🔗 utils.py	
> sac	
> utils	
🔗 __init__.py	
> safe_rl.egg-info	
▼ scripts	
🔗 experiment.py	M
🔗 plot.py	
🔗 test_policy.py	
🔗 .gitignore	
🔗 LICENSE	
🔗 README.md	
🔗 setup.py	M

3. Future works



Generalized coordinates: x (or θ)

3. Future works



?

?